Machine Learning Project — Topic Selection (Final Project)

CLINICPROOF™ — Sensor-Verified Medical Procedure Compliance

Course: ITAI 1371 — Introduction to Machine Learning, Spring 2026

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> **Note on distinctness from the midterm.** My midterm project was FIELDPROOF™, an oil-and-gas operations system for verifying field worker tasks using IMU and proximity sensors. The final project, CLINICPROOF™, is in healthcare and uses a different dataset (UCI HAR), a different problem framing, and a different evaluation context. The two share a common technical lineage — body-worn sensor data and supervised classification — but the application domain, regulatory framing, and ML-comparison plan differ enough to be graded as separate work and to strengthen my portfolio. A short comparison table appears at the end of this document.

Interest Area

Healthcare AI — using wearable sensors and machine learning to verify that clinical safety procedures are actually performed correctly.

Step 1 — Start With You

Pick one area you care about

Health / fitness — specifically the healthcare slice. I picked this for two reasons. First, healthcare is the domain where ML failure costs the most — a wrong answer in a movie recommender is forgettable; a wrong answer in a clinical compliance check can hurt a patient. Working on a problem where the stakes are real keeps me honest about evaluation, especially around false negatives and per-class performance. Second, my midterm project (FIELDPROOF™) was in industrial operations and used the same general technical approach — body-worn sensors, supervised classification of human activity. Moving the same kind of technical tooling into healthcare lets me reuse the lessons from the midterm while working on a much more meaningful problem.

Step 2 — Turn Your Interest Into ML Project Ideas

Healthcare is a huge field, so I focused on one specific failure mode: the gap between procedures that are documented as having happened and procedures that actually happened correctly. Three project ideas in the format Step 2 asks for:

1. Prediction framing. “Can ML predict whether a clinical procedure was performed correctly using wearable IMU sensor data?”

2. Classification framing — the one I am proposing. “Can ML classify human activity patterns into discrete categories (walking, sitting, standing, transitioning between rooms) using accelerometer and gyroscope signals?”

3. Detection framing. “Can I detect missed or skipped procedural steps from a sequential stream of body-worn sensor data?”

All three sit on the same continuum. The classification framing is the one I am actually building because it fits this course's scope: a public dataset, course-level supervised models, and metrics I can compare against published benchmarks. The prediction framing needs labeled clinical data that doesn't exist publicly, and the detection framing needs sequential modeling (LSTMs, transformers) that goes beyond what we've covered. The other two stay in the “future work” section of the final report rather than getting promised here and abandoned later.

Step 3 — Learn What the Best Work is Doing (State-of-the-Art)

Source 1 — Research paper

Citation: Anguita, D., Ghio, A., Oneto, L., Parra, X., & Reyes-Ortiz, J. L. (2013). *A Public Domain Dataset for Human Activity Recognition Using Smartphones.* Proceedings of ESANN 2013.

1. What are they trying to solve?

Whether smartphone-grade accelerometer and gyroscope signals are enough to classify everyday human activities accurately enough to be useful in real applications.

2. What data did they use? Can I access it?

They built and released the UCI Human Activity Recognition (HAR) dataset — 10,299 windowed sensor recordings from 30 volunteers wearing a Samsung Galaxy S II at the waist. The dataset is fully public on the UCI Machine Learning Repository, free to download, and is the dataset my project will use.

3. What ML method did they use?

A multiclass Support Vector Machine trained on a 561-feature engineered representation of the raw signals — time-domain statistics plus FFT-based frequency-domain features.

4. What's one important result?

Their tuned SVM hit roughly 96% test-set accuracy across the six activity classes, establishing the 89–96% benchmark band that follow-on papers compare themselves against.

Source 2 — Trusted article / industry source

Citation: The Joint Commission. (ongoing). *National Patient Safety Goals — Hospital Program.* Available at jointcommission.org/standards/national-patient-safety-goals/. The U.S. healthcare accreditation body whose published goals shape hospital practice.

1. What are they trying to solve?

Hospital-acquired infections, medication errors, wrong-site surgery, and other preventable harms — the same set of failures CLINICPROOF targets.

2. What data did they use? Can I access it?

They publish aggregate compliance and harm statistics drawn from accredited U.S. hospitals. Headline statistics are public, but the underlying audit-level data is not — which is part of the gap CLINICPROOF is designed to close.

3. What ML method did they use?

The Joint Commission itself doesn't deploy ML; it sets the protocol standards (WHO 6-step handwash, the Universal Protocol for surgical safety, the Five Rights of medication administration) that ML systems like CLINICPROOF would automatically verify against.

4. What's one important result?

Their reporting consistently shows hand-hygiene compliance sitting between 40 and 60 percent across U.S. hospitals despite decades of awareness campaigns — strong evidence that human-only verification has a ceiling and that automated, sensor-based verification is the natural next step.

Step 4 — Look at 3 Big Companies

I picked three companies that each occupy a different slice of the sensor-plus-ML-in-healthcare space. Together they triangulate where CLINICPROOF would compete and where it would be genuinely new.

Company 1 — Medtronic

1. Company: Medtronic — one of the largest medical-device companies in the world, and a company I admire because its products touch patient outcomes directly.

2. Product / technology: The GI Genius intelligent endoscopy module is their flagship recent ML product, using computer vision in real time to highlight potential polyps during a colonoscopy.

3. How does ML help them? A deep-learning detector running on endoscopy video flags suspicious tissue regions faster than a human operator can scan, raising adenoma detection rates in published trials.

4. One interesting detail I learned: Medtronic didn't train the model in-house from scratch — GI Genius came out of an acquisition (Cosmo Pharmaceuticals' AI subsidiary). It's a useful reminder that even the biggest medical-device players buy ML capability rather than build it, which says something about how valuable applied-ML talent is in this industry.

Company 2 — Microsoft (Microsoft Health Futures / Microsoft for Healthcare)

1. Company: Microsoft, specifically its healthcare-focused research and product groups.

2. Product / technology: A range of products aimed at clinical documentation and workflow — Dragon Ambient eXperience (DAX) Copilot, Azure Health Bot, Nuance integrations — most of which reduce the manual documentation burden on clinicians.

3. How does ML help them? Speech recognition, large-language-model summarization, and structured-data extraction turn an unstructured doctor-patient conversation into a clinical note that drops directly into the EHR. That's a different angle on the same problem CLINICPROOF addresses (manual documentation is unreliable), but they attack it from the language side rather than the activity side.

4. One interesting detail I learned: Microsoft's clinical-documentation products explicitly position themselves around “reducing clinician burnout,” not just “automating tasks” — that's a useful framing template for CLINICPROOF since automating compliance verification reduces the same kind of administrative exhaustion.

Company 3 — Apple (Health & Research)

1. Company: Apple — through Apple Watch and Apple Health, the most consumer-visible player putting medical-grade biosensing on a wrist.

2. Product / technology: The Apple Watch sensor stack — accelerometer, gyroscope, optical heart sensor, ECG, blood-oxygen — paired with on-device ML models for fall detection, atrial-fibrillation notification, and activity classification.

3. How does ML help them? Classifiers run continuously on the watch's sensor stream to recognize specific events (a hard fall, an irregular heart rhythm, a workout starting) and surface them to the user or to clinicians via Apple Health.

4. One interesting detail I learned: Apple's fall-detection model was trained on data from older adults specifically performing stunt-coordinator-staged falls — in other words, the realistic-failure data they couldn't collect in the wild was created on purpose. That's the same data-collection problem CLINICPROOF will eventually face for “procedure done wrong” data, and it's good to know the playbook for solving it exists.

Step 5 — Shrink Your Topic to Match This Course

The full CLINICPROOF vision is a multi-system platform — wearables, environmental sensors, ML inference, and a blockchain compliance ledger — and is much too big for one semester. The course-scoped version answers a single question: **can a classical ML pipeline trained on public IMU sensor data classify human activities at the accuracy a healthcare safety system would require?** If yes, the bridge to clinical compliance is data collection and protocol mapping, not algorithmic invention.

1. Dataset.

UCI Human Activity Recognition Using Smartphones (Anguita et al., 2013), downloaded from the UCI Machine Learning Repository. 10,299 windowed sensor recordings from 30 volunteers, pre-split into 7,352 training samples and 2,947 test samples (subject-aware split — 21 subjects in train, 9 in test). Public, free, well-documented, with decades of published benchmarks I can compare against.

2. Input (features).

A 561-dimensional feature vector per recording window, derived from the smartphone's accelerometer and gyroscope signals (50 Hz sampling, 2.56-second windows with 50% overlap). Features include time-domain statistics (mean, std, max, min, IQR, energy, entropy), frequency-domain features from FFT (mean frequency, spectral energy, spectral entropy), and inter-axis correlations. Pre-normalized to the range [-1, 1].

3. Output (target).

One of six activity classes — WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING. In the clinical mapping I lay out in the abstract, WALKING corresponds to a healthcare worker moving between rooms, SITTING and STANDING to stationary procedural work, and the transitions to task initiation and completion.

4. Algorithm type.

Supervised multi-class classification. Each input window has a known activity label, and the model learns the mapping between sensor patterns and labels.

5. Models I will try.

Four classical models, chosen to span the bias-variance spectrum so the comparison maps directly onto Modules 8 and 9 of the course. Each one is intentional, not filler:

- **Logistic Regression** — the linear baseline. Tells me whether the problem is even hard. If it does well, the problem is closer to linearly separable than I'd guess.
- **K-Nearest Neighbors (k = 7)** — distance-based, non-parametric. The pre-normalized features remove the main thing that breaks KNN, so this is a fair contrast to the linear baseline.
- **Linear Support Vector Machine (C = 10)** — the historical strong baseline on UCI HAR. Maximizing the margin between classes is the right inductive bias for this kind of feature space.

- **Random Forest (n = 200)** — bagging ensemble. Variance reduction on top of decision trees, plus free feature importances, plus the most robust default I know on tabular data of this size.

Comparison with the Anguita et al. paper from Step 3: their published benchmark is roughly 96% accuracy with a tuned multiclass SVM. My target is to land inside the 89–96% benchmark band, with Random Forest as the most likely best performer in my comparison and Linear SVM in second place — a result that would replicate the paper at lower complexity and confirm the feasibility piece CLINICPROOF needs.

Step 6 — Abstract

CLINICPROOF is a sensor-verified medical procedure compliance system whose final-project scope is to demonstrate that classical machine learning, trained on public human activity recognition data, can classify body-movement patterns at the accuracy a healthcare safety system would demand. **The problem:** the World Health Organization estimates that one in ten patients is harmed during hospital care, and much of that harm comes from procedures that are documented as having happened but are not actually performed correctly — hand-hygiene compliance sits at 40–60% against a 95% target, and medication-administration errors kill thousands of patients annually. **Why it matters:** the verification layer in healthcare is still mostly manual paperwork, and that paperwork is slow, inconsistent, and falsifiable; replacing it with a sensor-verified record would directly reduce infections, medication errors, and malpractice exposure. **State of the art:** Anguita et al. (2013) established that smartphone IMU data can classify daily activities at 89–96% accuracy with a tuned SVM, and the field has matured since then; meanwhile The Joint Commission's compliance reporting documents the human-verification ceiling that automated approaches need to break. **Industry:** Medtronic deploys ML in surgical workflows (GI Genius polyp detection), Microsoft automates clinical documentation (DAX Copilot), and Apple has put medical-grade activity classification on a consumer wrist — but none of them tie sensor-based activity recognition to specific clinical-protocol verification, which is the gap CLINICPROOF would occupy. **What I will build:** a complete supervised-classification pipeline in Google Colab using the UCI HAR dataset (10,299 windowed samples, 561 features, six activity classes), comparing Logistic Regression, K-Nearest Neighbors, Linear SVM, and Random Forest with full evaluation including per-class metrics, confusion matrices, and feature-importance analysis. The success criterion is at least 90% test-set accuracy and below 5% false-negative rate on movement-related classes — the threshold a clinical safety system would require — with results compared directly against the Anguita et al. benchmark.

Step 7 — Talk About Your Idea

This submission contains everything Step 7 asks for:

- The interest area as a one-sentence topic statement (top of this document).
- Steps 1–6 as Heading-1 sections, each guideline question as a Heading-2 sub-question with an answer underneath.
- A one-paragraph abstract under Step 6 covering the five required elements (problem, why it matters, state of the art, industry, and what I will build).

Five-minute talk plan

Roughly how I'd walk through this in a five-minute presentation:

• **00:00–00:45 — Why this matters.** The WHO 1-in-10 patient-harm number, the 40–60% hand-hygiene gap, and the basic claim that the healthcare verification layer is still manual and falsifiable. The "why anyone should care" section.

• **00:45–01:45 — The CLINICPROOF concept.** Wearable IMUs plus environmental sensors plus ML, producing an immutable compliance record that replaces paper. One sentence on each of the four use cases (hand hygiene, medication administration, surgical safety, fall prevention) so the audience sees the breadth.

• **01:45–02:30 — Scoping the project.** Why I'm only solving the ML feasibility core, not the whole platform — and why UCI HAR is the right stand-in dataset for demonstrating that the underlying technology works.

• **02:30–04:00 — The technical plan.** Four models, why each one, what the comparison will show, and the 90% / <5% FN success criterion. The section where the bias-variance arc from Modules 8 and 9 gets made concrete.

• **04:00–04:30 — Distinctness from FIELDPROOF.** Yes the sensor-classification lineage is shared, no the project is not the same: different domain, different dataset, different evaluation framing. (Comparison table at the end of the document for reference.)

• **04:30–05:00 — Honest limitations and Q&A handoff.** The dataset is a stand-in; six classes is a small ontology; sequential modeling is out of scope; per-subject generalization is the real test. Most likely audience question is "why not deep learning?" — the answer is interpretability matters more than the last 1–2 percent of accuracy in a clinical system.

Appendix — Distinctness from Midterm Project (FIELDPROOF™)

Per the guideline note that "the two topics and projects should be sufficiently distinct," here is the side-by-side. The two share a common technical lineage — body-worn sensors and supervised classification — which is deliberate (it lets me build on what I learned), but the application domain, dataset, evaluation framing, and ML comparison plan are all different.

Dimension	Midterm — FIELDPROOF™	Final — CLINICPROOF™
Domain	Oil & gas field operations	Hospital procedure compliance
Problem	Verify task execution by field workers	Verify clinical procedure compliance by healthcare workers
Stakes	Operational safety, regulatory audits	Patient safety, infection control, malpractice
Dataset	Synthetic / FIELDPROOF lab data	UCI HAR (10,299 samples, 561 features, public)
Sensors	IMU + proximity (lab simulated)	Smartphone IMU at the waist (real volunteers)
Classes	Task-execution outcomes	Six activity types (WALKING, SITTING, …)
Models compared	Course models on midterm-specific data	Logistic Regression, KNN, Linear SVM, Random Forest
Benchmark	Internal lab targets	Published 89–96% accuracy band (Anguita et al., 2013)
Evaluation focus	Operational reliability	False-negative rate < 5% (clinical-safety floor)

Bottom line: FIELDPROOF taught me how to wire body-worn sensor data through a supervised classification pipeline. CLINICPROOF takes that capability and applies it to a different domain, a different dataset, and a different success criterion — with the explicit goal of demonstrating the technical feasibility step that a real clinical compliance system would have to pass before any deployment.